

Layers Chosen:

Three data layers were selected and applied to the base map for our analysis:

1. Provinces and Territories Layer:

- **Geometry Data Type:** Polygons representing the boundaries of provinces and territories in Canada (with the Definition Query set on New Brunswick).
- **Attribute Data:** Each polygon here represents attributes like population, area size, and province name.

Spatial Reference: The 'WGS 1984 Web Mercator (auxiliary sphere)' Projected Coordinate System was used here to ensure the geographic coordinates are perfect for the map's scale.

2. Sustainable Transportation Layer:

- **Geometry Data Type:** Census division polygons (also represented with colour coding) that shows the proportion of commuters using sustainable transportation.
- **Attribute Data:** This includes the percentage of commuters using sustainable transportation (like cycling, walking, or public transport) based on the 2016 census data, categorized into ranges (legend).

Spatial Reference: The 'NAD 1983 Canada Atlas Lambert' Projected Coordinate System was used here to ensure correct analysis at the census division level. The polygons used adequately visualize these divisions.

1. Greenhouse Gas Emission from Large Facilities Layer:

- **Geometry Data Type:** Point and Polygon.
This layer represents the emissions using points and shapes(polygon) to mark the locations of extensive facilities emitting greenhouse gases.
- **Attribute Data:** Each point represents data attributes detailing the volume of greenhouse gases emitted (in kilotons of CO equivalents) by each facility and industry type, categorized into different levels/ranges.

Spatial Reference: The 'NAD 1983 Canada Atlas Lambert' Projected Coordinate System was used here to ensure accurate placement of these facilities by aligning with the base map's projection.